

Technical Document #857: Retrieval Augmented Generation - Comprehensive Overview

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Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Hybrid search combines keyword-based and semantic search.
- Query decomposition breaks complex queries into simpler sub-queries.